

Deep Learning for Computer Vision

114-1/Fall 2025

<https://cool.ntu.edu.tw/courses/53063> (NTU COOL)

<http://vllab.ee.ntu.edu.tw/dlcv.html> (Public website)

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Deep Learning for Computer Vision

- Time:
 - Wednesday 2, 3, 4
 - ~~09:10-12:20~~ → **09:30-12:20** with 1 short break
- Location:
 - BL-113
- Websites:
 - NTU COOL; primary website: <https://cool.ntu.edu.tw/courses/53063>
 - Public use: <http://vllab.ee.ntu.edu.tw/dlcv.html>

Deep Learning for Computer Vision (cont'd)

- **Required** Knowledge & Skills
 - Knowledge of **linear algebra, vector calculus, and probability**
 - **Machine learning & deep learning** (this is **NOT** an entry-level course)
 - **Programming skills** (in Python)
 - *If any of the above skills not available, please come back when ready.*
- This course is offered in **Mandarin** this semester.
Q&As & discussions can be done in any languages communicable.
- To encourage class attendance, lectures will be recorded but will **NOT** be uploaded to COOL **until a week after**.

What to Expect from this Course?

- (Deep) Learning-Based Computer Vision
 - ~~Fundamentals of machine learning~~
 - **Deep learning** technologies for visual classification, synthesis, and beyond (e.g., LLM & foundation models)
 - Different yet relevant courses at NTU: like *Computer Vision*, *3D Vision*, *Applied Deep Learning*, *RL*, etc.
- Practical Experiences
 - Assignments and projects dealing with real-world visual data & problems
 - Final projects associated with challenges from CVPR, ICCV, etc.
 - Possibly come w/ cash prizes for top teams
- **Lots of work** with fast pace, but hopefully rewarding & useful!



Disclaimer

- You are responsible for **your own computing resources**.
(e.g., GPUs, Colab Pro < \$30/month, etc.)
You **can't** finish this course w/o using GPUs.
- While doing my best to stick to what we announce,
course materials (e.g., syllabus, HW policy, etc.) might change over time.
- **Up to 20% 33% of students** chose to drop the course last year.
- And, yes, **we did fail students in the past years (~5% each semester)**.



Required Computing Resources

- If you're using your own machines...(update by HW#1)
 - Ubuntu 22.04.4 LTS
 - NVIDIA RTX A4500 (20GB)
 - GNU bash, ver. 5.1.16(1)-release
 - Python 3.8

Course Information

- Teaching Team & Office Hours
- Course Policy
- How to enroll in this class if not already in?

About Myself

- **Education**



- **Ph.D./M.S. in Electrical & Computer Engineering** 2002 – 2009
Carnegie Mellon University, Pittsburgh, USA



- **B.S. in Electrical Engineering** 1997 – 2001
National Taiwan University, Taipei, Taiwan

- **Experiences**



- **NVIDIA** AI研究總監, NVIDIA Research 2022/08 – present

- **Professor**, Dept. EE, National Taiwan University 2019 – present



- **Principal AI Consultant**, Inventec 2021 – 2022

AICS



- **AI Advisory Consultant**, ASUS Intel. Cloud Services (AICS) 2019 – 2022

- **Associate Professor**, National Taiwan University 2017 – 2019

- **Deputy Director**, CITI, Academia Sinica 2015 – 2017

- **Associate/Assistant Research Fellow** 2009 – 2017

Research Center for IT Innovation (CITI), Academia Sinica



- **Office Hour**

- After class or by appointment via email (ycwang@ntu.edu.tw)

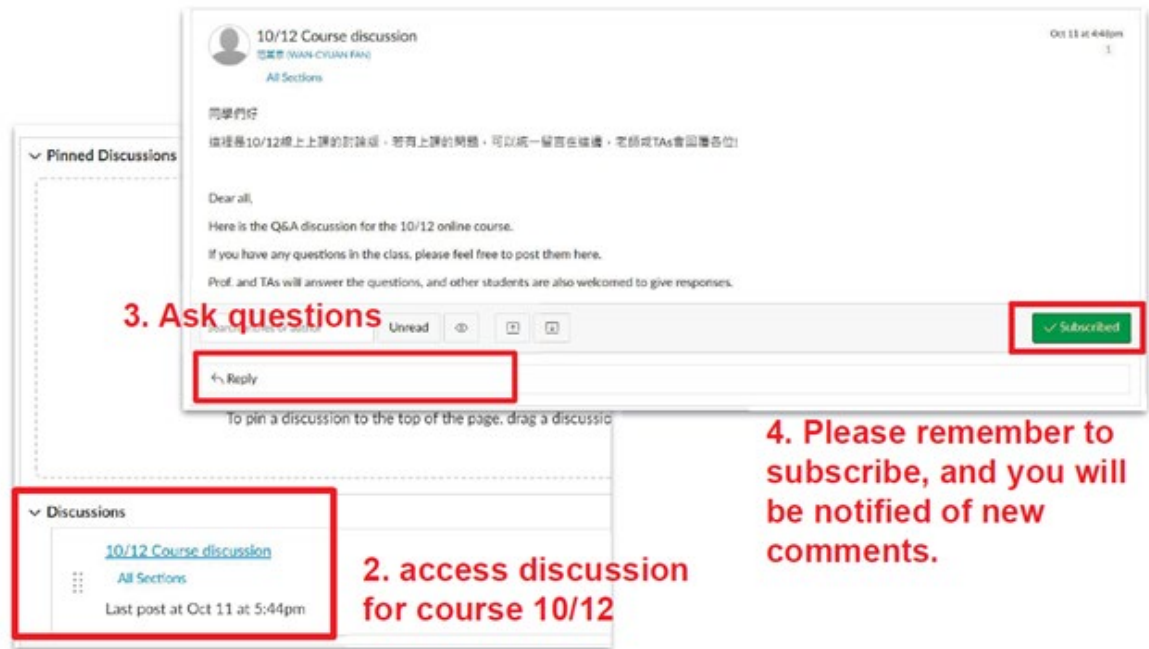
TAs & Office Hours

- We have **>8** TAs this semester.
 - Will post and update the TA info on DLCV website
- If any questions, you are encouraged to
 - Ask at NTU COOL (anonymous is fine)
 - Contact your TA **by email at** ntudlcv@gmail.com
- Q&A @ NTU COOL
 - “Discussions” -> “Q&A for Date” -> Ask your questions @ Reply

The screenshot displays the NTU COOL website interface. On the left, a sidebar menu for the course '深度學習於電腦視覺 (COMME5052)' includes links for Home, Syllabus, Modules, Announcements, Assignments, Quizzes, Discussions, and Grades. The 'Discussions' link is highlighted with a red box and labeled '1. click discussions'. The main content area shows a 'Pinned Discussions' section with a link to '10/12 Course discussion' highlighted by a red box and labeled '2. access discussion for course 10/12'. To the right, the discussion page for '10/12 Course discussion' is shown, with a 'Subscribed' button highlighted by a red box and labeled '4. Please remember to subscribe, and you will be notified of new comments.' Below the discussion title, a 'Reply' button is highlighted by a red box and labeled '3. Ask questions'.

Q&A @ NTU COOL

- “Discussions” -> “Q&A for Date” -> Ask your questions @ Reply

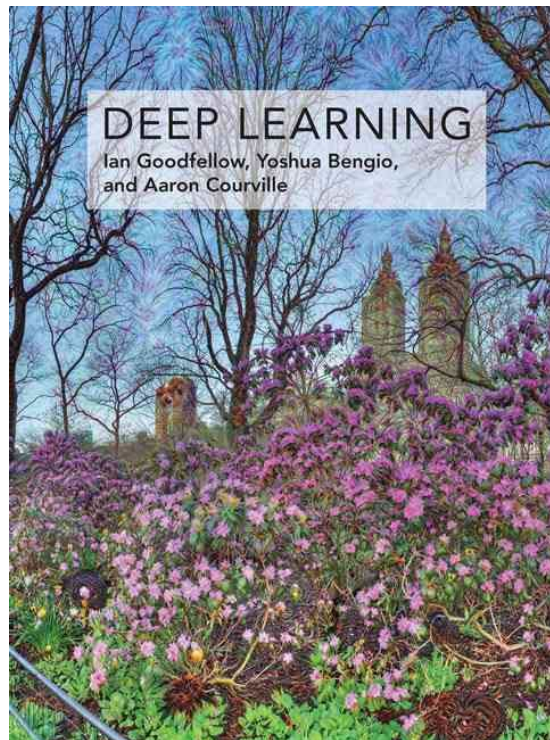


Syllabus

Week	Date	Topic	Course Materials	Remarks
1	09/03	Course Logistics; Intro to Neural Nets		
2	09/10	Convolutional Neural Networks & Self-Supervised Learning		HW #1 out
3	09/17	Image Segmentation & Object Detection		
4	09/24	Generative Models (I) - AE, VAE, & Diffusion Model (I)		HW #1 due
5	10/01	Generative Models (II) - Diffusion Model (II), GAN		HW # 2 out
6	10/8	Recurrent Neural Networks & Transformer (I)		
7	10/15	Guest Lecture (TBD)		HW #2 due; ICCV week
8	10/22	Transformer (II); Vision & Language Models; Reasoning		HW #3 out
9	10/29	Parameter-Efficient Finetuning; VLM Personalization & Editing		
10	11/05	Guest Lecture: Toward Efficient LLM Inference - Challenges & Solutions by Prof. Kai-Chiang Wu, NYCU		
11	11/12	Guest Lecture: 3D Vision by Dr. Sheng-Yu Huang, NTU/NVIDIA		HW #3 due HW #4 out
12	11/19	Guest Lecture: Linda Huang, Senior Dir., GeValyn Associates		
13	11/26	Guest Lecture: Vision Language Action Models (VLA) by Dr. Fu-En Yang, NVIDIA		Final Project Announcement
14	12/03	Guest Lecture (TBD)		HW #4 due; NeurIPS week
15	12/10	Advanced topics in Foundation & World Models		
17	12/23 Tue	Final Project Presentation		

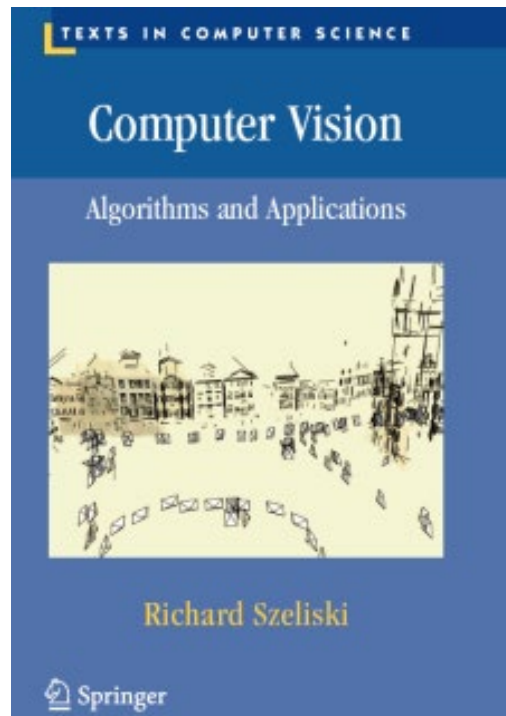
Textbook (Optional)

- Deep Learning, MIT Press
 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville
 - Free online versions available at www.deeplearningbook.org

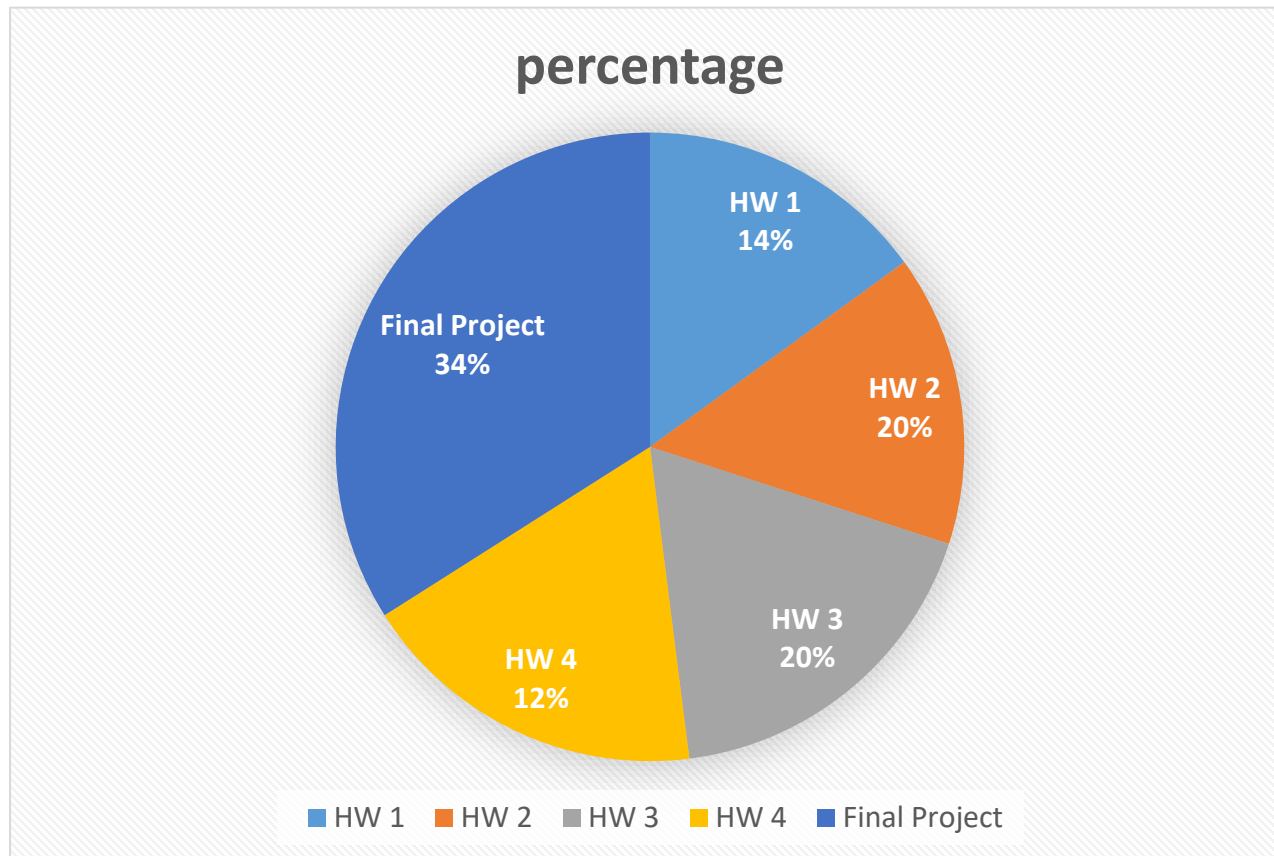


Textbook (Optional)

- Computer Vision: Algorithms and Applications, Springer
 - Richard Szeliski
 - Free online versions available at <http://szeliski.org/Book/>



About Grading



About Grading (cont'd)

- HW assignments **(66%)**
 - ~~HW 0 (non-DL HW, 0% but required)~~
 - HWs #1~4 (12~20% each with possible bonus points)
 - ~~HW #5 (optional yet with bonus points)~~
- Final project + poster presentation + code, etc. **(34%)**
- **Bonus points**
 - Course participation (e.g., interaction, Q&A, etc.)
 - Extra challenges in HWs
 - Excellent performance for final project (e.g., ranking, presentation, publication submissions, etc.)



Final Grade

Letter Grading System	Definition	Grade Points	Conversion Scale
A+	All goals achieved beyond expectation	4.3	90-100
A	All goals achieved	4.0	85-89
A-	All goals achieved, but need some polish	3.7	80-84
B+	Some goals well achieved	3.3	77-79
B	Some goals adequately achieved	3.0	73-76
B- (passing grade for graduate students)	Some goals achieved with minor flaws	2.7	70-72
C+	Minimum goals achieved	2.3	67-69
C	Minimum goals achieved with minor flaws	2.0	63-66
C- (passing grade for undergraduate students)	Minimum goals achieved with major flaws	1.7	60-62
F	Minimum goals not achieved	0	59 and below
X	Not graded due to unexcused absences or other reasons	0	0
W	Withdrawal		
NG	No grade reported		
IP	In progress		
TR	Transfer credit		
EX	Exempted		

About Course HWs

- About HW late policy
 - We offer free late days (up to **THREE** days) in case you have dates, midterms, HW dues for other course, etc. (如期中考、專題、社團、約會、找不到人抄)
 - 1 min ~ 23 hr 59 min all count as ONE late day.
 - After HW due day, a penalty of **30%** per day.
 - We'll maximize your final score based on HW scores and the late days used.
 - **No** late submission for the final project, obviously.

About Course Projects

- About Final Project

- Details to be announced around mid semester
- 3~4 people per group (no more than 4)
- TAs will check your code after presentation
- Selected topics possibly come with cash prizes.
- Evaluated by instructor, TAs, and possibly guest judges
- (Intra/inter-group) peer evaluation will be conducted.
- Snack/drinks will be provided during final presentation.



Academic Integrity

- Can discuss HW with peers, but DO NOT copy and/or share code
 - Plagiarism is against university policy.
 - Violation in ANY form for HWs & final project would result in F.
 - We gave at least five Fs in previous semesters for the above cases.
- Do not directly use code/results from Internet unless you have permissions.
 - If not sure, ask!
 - If so, do specify in your HW/project.
- No double dipping!
 - You **CANNOT** use your lab research work as your final project.
 - However, you are encouraged to extend your previous work.
 - Some students successfully turned their final projects into international publications in the past semesters!

DOs and DONTs for the TAs (& Instructor)

- Do NOT send **private** messages to the TAs via Facebook, etc.
 - TAs are here to help, but they are not your tutors 24/7.
- TAs will **NOT** debug for you, including coding, environmental, library dependency problems.
- If you cannot make the TA hours, please email & schedule an appointment
- Obviously, TAs do **NOT** answer questions not related to the course.
- You are encouraged to discuss w/ me if any questions on (but not limited to) **DLCV, research, career planning**, etc.

How to Sign Up If Not Already In?

- **Capacity**

- Classroom capacity: 140; currently registered: 100
- About **20** students can be additionally added.
- If you do not plan to take the course anymore, please drop it ASAP.

- **Priority**

- EECS students > students with strong research needs (w/ justifications) > those are interested in AI/GenAI...
- If you are still interested in this course and plan to enroll, please fill in the following form **Sept 3rd Tue between 9am-2pm:**
<https://shorturl.at/FEDs0>
- We will announce the enrollment results via email no later than **Sept 6th Sat 12pm. All decisions are final.**





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Any Questions?

Course registration at <https://shorturl.at/FEDs0>
by **Sept. 3rd Wed 2pm!**